Black rice anthocyanins inhibit cancer cells invasion via repressions of MMPs and u-PA expression.

Tumor metastasis is the most important cause of cancer death and various treatment strategies have targeted on preventing the occurrence of metastasis. Anthocyanins are natural colorants belonging to the flavonoid family, and are wildly used for their antioxidant properties. Here, we provided molecular evidence associated with the anti-metastatic effects of peonidin 3-glucoside and cyanidin 3-glucoside, major anthocyanins extracted from black rice (Oryza sativa L. indica), by showing a marked inhibition on the invasion and motility of SKHep-1 cells. This effect was associated with a reduced expression of matrix metalloproteinase (MMP)-9 and urokinase-type plasminogen activator (u-PA). Peonidin 3-glucoside and cyanidin 3-glucoside also exerted an inhibitory effect on the DNA binding activity and the nuclear translocation of AP-1. Furthermore, these compounds also exerted an inhibitory effect of cell invasion on various cancer cells (SCC-4, Huh-7, and HeLa). Finally, anthocyanins from O. sativa L. indica (OAs) were evidenced by its inhibition on the growth of SKHep-1 cells in vivo.